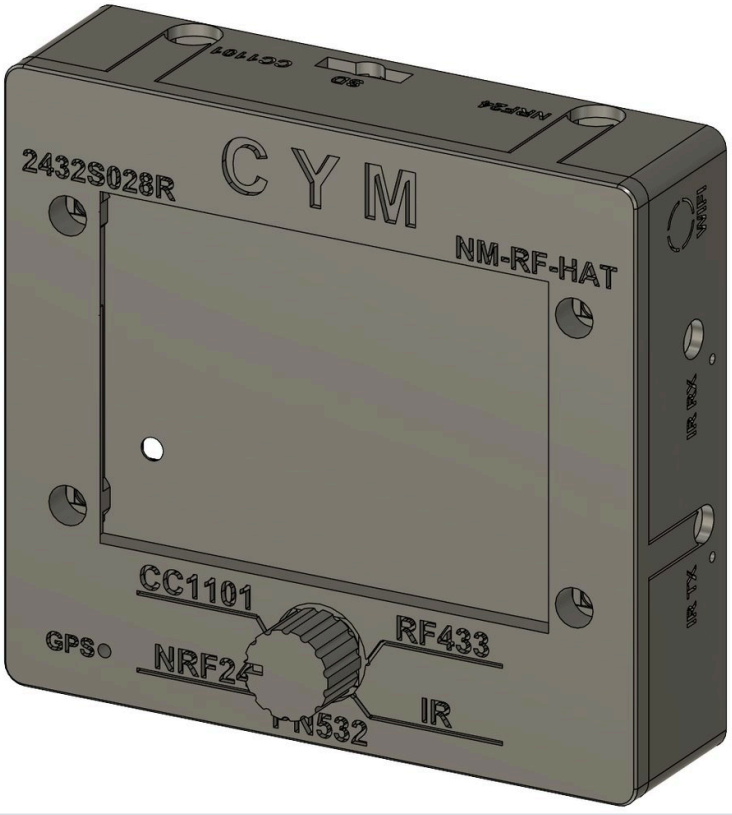
CYM Case 2432S028R RF-HAT – 3D Printing and Installation Manual

Case with battery, GPS and rotary module selector for the ESP32-2432S028R (CYD) + NM-RF-HAT · Revision 2

Designed and made by @birolt29 @alen29





The real device (CYM interface running) and the 3D design view.

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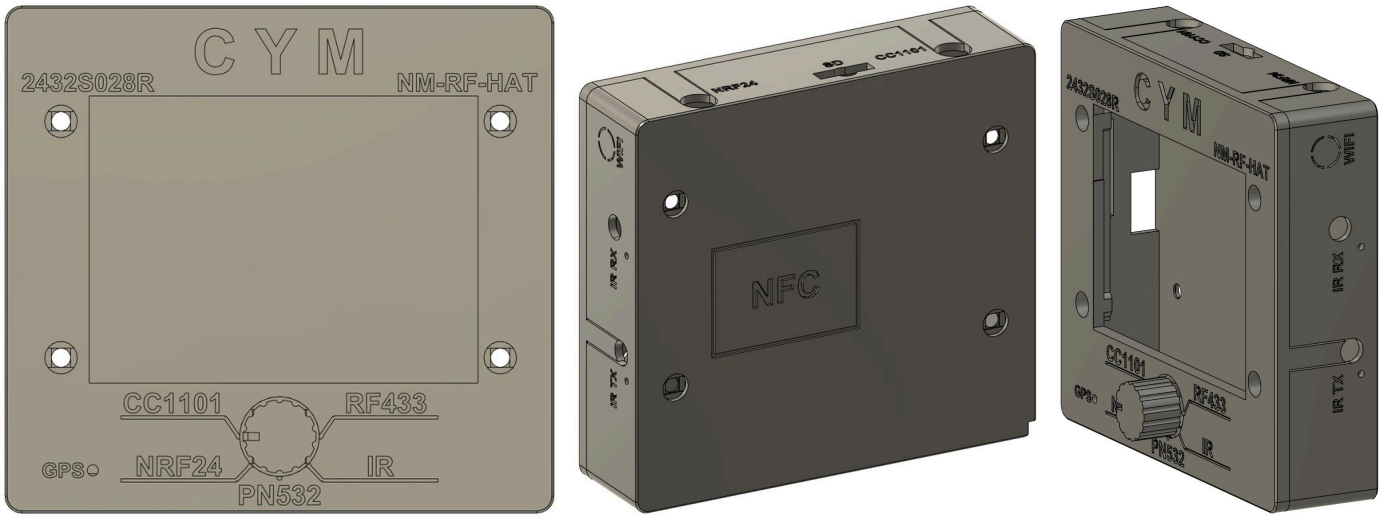
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This manual has four parts: general information about the boards and the parts used, 3D printing of the case, the front cover and the knob, assembling the parts into the case, and installing the firmware. Print settings are given by their generic names; you will find the equivalents in whichever slicer you use.

The case holds the ESP32-2432S028R main board (front) and the NM-RF-HAT expansion board (back) together; the two boards are stacked on the 15 mm brass standoffs from the set, with the battery between them. The HAT's 6-way DIP switch is replaced by a **5-position rotary switch** on the front (module selection) and a **slide switch** on the side (power). The GPS module and its antenna sit in pockets inside the front cover, the stylus in a channel along the bottom edge. The back wall over the NFC reading area is kept thin.



Finished device: front (screen on), back (NFC area) and antennas, left wall (USB, CHARGE, I/O).



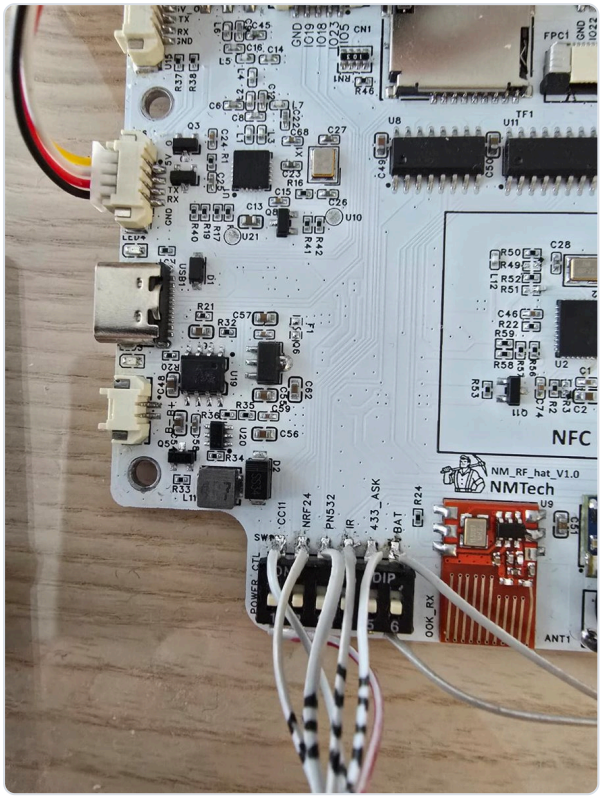
Design views: front, back, right wall (WIFI, IR RX, IR TX).

Part 1 – General information

Board specifications

Main board	ESP32-2432S028R (CYD) · ESP32-WROOM-32, dual-core, 240 MHz · 4 MB flash · Wi-Fi 2.4 GHz, Bluetooth 4.2
Display	2.8" TFT · 320×240 px · ILI9341 · resistive touch (XPT2046), stylus included
Expansion board	NM-RF-HAT (NMTech): CC1101 433 MHz (SMA) · nRF24L01 2.4 GHz (SMA) · PN532 NFC 13.56 MHz · RF433 OOK/ASK transceiver · IR transmitter and receiver · microSD · GNSS socket
Power	Through the HAT: USB-C (5 V, 500 mA charging) or a 3.7 V LiPo (MX1.25 socket, 2.5–4.2 V); the HAT powers the CYD through its P1 socket · total draw 1–1.9 W
Module selection	Only one module is on at a time. In this case with the 5-position rotary switch: CC1101 · NRF24 · PN532 · IR · RF433
Case dimensions	97.5 × 90.1 mm · total thickness about 27 mm (case 22.15 + front cover 5.5; knob not included)

Connectors (pinout)



Socket	Pins	Connects to
HAT IN	1 5V · 2 TX · 3 RX · 4 GND	CYD P1 (1 VIN · 2 TX · 3 RX · 4 GND) – 4-wire cable from the HAT kit, pin to pin (1→1 … 4→4); the CYD is powered through it
HAT I2C	1 3.3V · 2 IO27 · 3 IO22 · 4 GND	CYD CN1 (1 3.3V · 2 IO27 · 3 IO22 · 4 GND) – 4-wire cable from the HAT kit, pin to pin
HAT TF 5P	GND · MISO · CLK · MOSI · CS	TF extension adapter (SD adapter, HAT kit) plugged into the CYD's microSD slot; the HAT's own SD slot is used
HAT OUT / GNSS	1 5V · 2 TX · 3 RX · 4 GND	GPS module ATGM336H: 5V → VCC, TX → RX, RX → TX, GND → GND (TX and RX crossed)
HAT BAT	B+ · B− (MX1,25)	LiPo battery, 3.7 V
HAT DIP pads	1 CC1101 · 2 NRF24 · 3 PN532 · 4 IR · 5 433_ASK · 6 BAT	1–5 to the rotary switch, 6 to the slide switch; the common/outer pin of each switch to GND (the other side of the DIP is common GND)

Photo: the side of the HAT that faces the CYD; the wires of the rotary switch and the slide switch are soldered to the DIP switch pads. The socket names are printed on the board. Wire colours can differ from kit to kit; go by the socket and pad names and check the battery positive with a multimeter.

Reference: parts and firmware used

Part	Product	Link
Main board	ESP32-2432S028R (CYD-2432S028), ESP32 with a 2.8" touch screen; its set includes screws, standoffs, a stylus and a 4-wire MX1.25 cable	www.amazon.de/dp/B0D2TN77LY
Expansion board	NMTech NM-RF-HAT; its kit includes 4 × 15 mm brass standoffs, screws, the CYD–HAT cables, the TF extension adapter (SD adapter) and two SMA antennas	aliexpress.com/item/1005010751235469
Battery	3.7 V 3000 mAh (11.1 Wh) LiPo with MX1.25 plug, 35.5 × 66.5 × 10.5 mm; labelled "MakerFocus"	www.amazon.de/dp/B0F18QM8SJ
GPS module	ATGM336H GPS+BDS Dual Mode Module	www.amazon.de/dp/B0FKH26344
Rotary switch	Batum 1-pole 5-position rotary switch (RS16 1P5T, pack of 4): body Ø16 mm, shaft 15 mm (incl. thread), M9 bushing, 45° steps, 0.3 A 60 V	www.amazon.de/dp/B0GYTJB4T1
Switch	SS12D00G slide switch, 3 mm lever ("12D00G-3mm"); from the VEXUNGA 80-piece slide switch set	www.amazon.de/dp/B0FG2TSLFC
Firmware	Cheap Yellow Monster (CYM)	github.com/JimGat/CYM#supported-hardware

Both boards come as kits. From the CYD kit you use the front cover screws, the stylus and the 4-wire MX1.25 cable for the GPS module; from the HAT kit the 4 × 15 mm brass standoffs, the back screws, the CYD–HAT cables, the TF extension adapter (SD adapter) and the two SMA antennas. The GPS kit contains only the module and its antenna. You also need the battery, the GPS module, the rotary switch, the slide switch and a thin 6-wire ribbon cable. The case has been tested with these parts. The battery sits between the two boards and has no pocket of its own: 3000 mAh and larger batteries fit, but before ordering check that it is no thicker than 10.5 mm and that its lead reaches the HAT's BAT socket. Links as of 16.10.2026; product pages may change. The firmware supports the board as "CYD-2432S028"; select it as the board on the installer page.

> Caution – battery polarity: with MX1.25 battery plugs the order of plus and minus differs between makers, and wire colours are not reliable either. Before plugging the battery into the HAT, check the positive terminal of the battery and of the HAT's BAT socket with a multimeter. Reversed polarity destroys the HAT's charging circuit.

Part 2 – 3D printing

Print files

File	Contents
`CYM_Case_2432S028R_RF-HAT.3mf`	All three parts together (case + front cover + knob), in print orientation. Use this if your slicer opens 3MF.
`CYM_Case_2432S028R_RF-HAT_Case.stl`	Case (body). Units: millimetres.
`CYM_Case_2432S028R_RF-HAT_Front_Cover.stl`	Front cover. Units: millimetres.
`CYM_Case_2432S028R_RF-HAT_Knob.stl`	Rotary switch knob. Units: millimetres.

Three parts are printed; no supports are needed. Case 97.5 × 90.1 × 22.15 mm, front cover 97.5 × 90.1 × 19.4 mm (including the tongues), knob Ø13 × 9.2 mm. All three together about 60 g of PLA.

> Revision 2 (16.10.2026): in the revision 1 files the stylus channel was closed by a thin wall about 3 mm inside the mouth; fixed. The SMA antenna holes were enlarged to Ø8 mm (the antenna nuts no longer bear on the case). A lip that narrows towards the front face was added to the left edge of the screen window; no light leaks past the side of the screen, and it does not touch the touch film strips. Do not use the old files.

Placing on the bed

- Case: the flat, closed back face (with NFC) goes on the bed; the open side faces up.
- Front cover: the front face with CYM goes on the bed; the tongues, the centering lip and the GPS pockets face up.
- Knob: its top face (marker line) goes on the bed; the shaft hole faces up.
- The files are saved in this orientation; do not rotate them. The bed edges have a 0.4 mm chamfer (elephant-foot allowance).
- Supports off. The counterbore roofs and the WIFI membrane print without supports thanks to bridging layers.

Material

PLA or PLA+ is enough. Choose temperatures from your filament label. Damp filament strings and fills in small lettering and hole edges; dry it before printing.

Slicer settings (0.4 mm nozzle)

Setting	Value	Why
Layer height	0.16 mm (first layer 0.20 mm)	Clean side lettering (2.2–2.4 mm) and 45° faces.
Wall (perimeter) count	4	1.8 mm walls print solid.
Infill	100%	No internal bridges (slow-downs) in thin plates; stronger parts.
Top / bottom solid layers	5 / 5	
Seam	Aligned	The seam does not land randomly on hole edges.
First layer speed / first layer infill	20 / 20 mm/s	The front cover lettering is on the bed face.

Setting	Value	Why
Elephant foot compensation	0.15 mm	Keeps the bed-face lettering and edge from closing up.
Bridge flow / bridge fan	0.9 / 100%	Counterbore roofs and engraved lettering roofs.
Outer / inner wall speed	60 / 150 mm/s	For a printer with calibrated input shaping (acceleration 4500, outer wall 1000). Otherwise outer wall 30 mm/s.
Supports	Off	

Setting names differ between programs (Cura, PrusaSlicer, OrcaSlicer, Bambu Studio, Creality Print etc.). Reference print: Ender-3 S1 Pro (Sonic Pad), textured PEI plate, 0.4 mm nozzle, PLA+, nozzle 210 / 200 °C, bed 60 / 55 °C; all three parts on one plate in about 3 hours 45 minutes. With these settings the screw holes and the rotary switch hole came out without blobs.

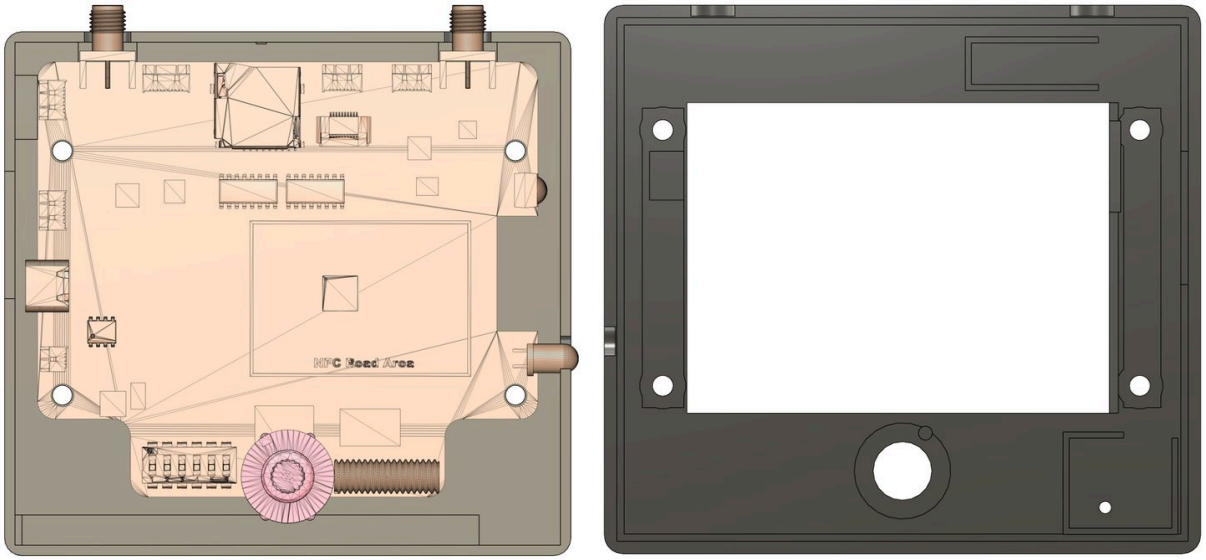
Checks after printing

- The thin bridging layers on the counterbore roofs (front: 4 screws + the rotary switch nut; back: 4 screws) are opened by pushing the screw through.
- The cover's three tongues (two SMA, IR TX) should go into the slots in the case without force; if tight, thin them slightly with fine sandpaper.
- Remove the WIFI membrane, by cutting its 4 bridges, only if you fit an external antenna.
- Are the HAT SD slot, the IR RX window and the USB and CHARGE openings clean? Any strings inside the engraved lettering?
- Are the GPS LED hole (front cover, next to the GPS label) and the rotary switch locking-pin hole (Ø2.3) open?
- The knob should fit the shaft snugly but without force (hole Ø6.1). The stylus channel must be open all the way.

Part 3 – Assembly

What you need

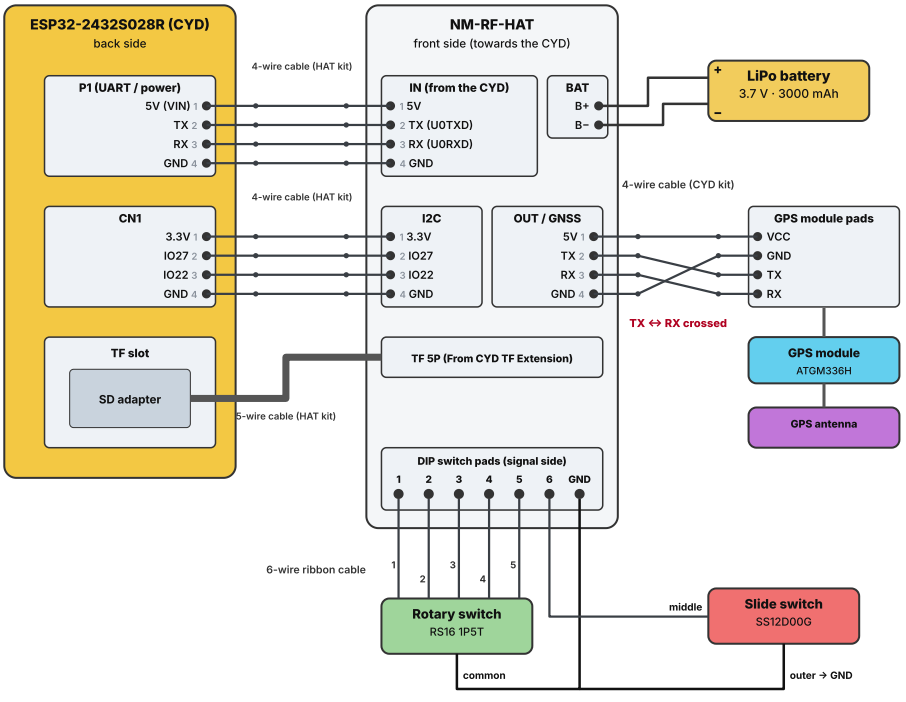
- Printed case, front cover and knob (CYM Case 2432S028R RF-HAT, revision 2)
- ESP32-2432S028R board and from its kit: 4 × M3 × 7 screws (for the front cover), the stylus and the 4-wire MX1.25 cable (for the GPS)
- NM-RF-HAT and from its kit: 4 × 15 mm brass standoffs (female both ends), 4 × M3 × 6 screws (for the back), two SMA antennas, two 4-wire cables, the TF extension adapter (SD adapter) and its 5-wire cable
- LiPo battery 3.7 V 3000 mAh, at most 10.5 mm thick, MX1.25 plug, protected
- GPS module ATGM336H (16 × 14 × 5.5 mm) and its antenna (19 × 6 × 5.5 mm, 90 mm cable)
- Rotary switch RS16 1P5T (with M9 nut and lock washer) and a thin 6-wire ribbon cable (≈15 cm)
- Slide switch SS12D00G, 3 mm lever, and 2 thin wires
- Thin double-sided tape, heat-shrink tubing, soldering kit, multimeter, side cutters, tweezers, hot glue



Layout (design): left, inside the case with the HAT in place (viewed from the front); right, inside of the front cover – pockets for the GPS module (bottom right) and antenna (top), the rotary switch seat, the tongues and the centering lip.

Wiring diagram

Which wire goes where: between the CYD and the HAT, the two 4-wire cables from the HAT set and the SD adapter's 5-wire cable; the battery to the HAT's BAT socket; the GPS module to the HAT's OUT / GNSS socket (TX and RX crossed); the rotary switch and the slide switch to the pads of the HAT's DIP switch. Wire colours can differ from kit to kit; connect by socket and pad names.



Grey numbers are pin numbers; the HAT kit cables connect pin 1 to pin 1. All 6 DIP levers stay OFF; module via the rotary, power via the slide switch. Each line is one wire; the names at its ends are the socket and pad names. Check the battery positive with a multimeter.

> Caution: keep the slide switch off (O) while flashing through the CYD's USB socket; the HAT powers the CYD through its P1 socket, so the two supplies must not be connected at the same time. The battery charges through the HAT's USB-C socket labelled CHARGE (500 mA); it also charges with the slide switch off, because the switch only sits between DIP 6 (BAT) and GND.

Rotary switch positions

Position	Front label	HAT pad	Module
1	CC1101	DIP 1	433 MHz transceiver (left SMA antenna)
2	NRF24	DIP 2	2.4 GHz transceiver (right SMA antenna)
3	PN532	DIP 3	NFC reader (NFC area on the back)
4	IR	DIP 4	IR transmitter (right wall IR TX) and receiver (IR RX)
5	RF433	DIP 5 (433_ASK)	433 MHz OOK/ASK transceiver (spring antenna on the HAT)

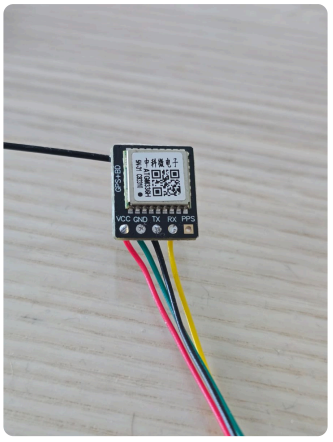
The rotary switch replaces levers 1–5 of the HAT's DIP switch: only the selected module is on at any time. After changing the module, open the matching menu in CYM; the GPS module is always powered, independently of the rotary switch.

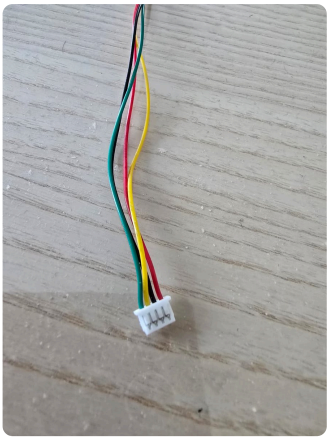
Preparation: soldering (at the table, before anything goes into the case)

Prep 1 Solder the cable to the GPS module

The plug end of the 4-wire MX1.25 cable from the CYD kit goes into the HAT's OUT / GNSS socket. Cut off the plugs at the other end and solder the wires straight to the GPS module pads (VCC, GND, TX, RX); a pin header does not fit the pocket in the front cover. Lead the wires off flat.

Connections: GNSS 5V → GPS VCC, GNSS TX → GPS RX, GNSS RX → GPS TX, GNSS GND → GPS GND; TX and RX are crossed. Wire colours can differ from kit to kit; go by the labels on the socket and on the module.



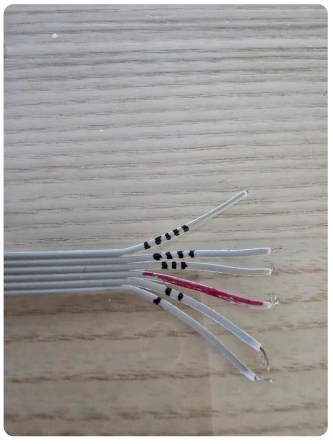


Prep 2 Prepare the rotary switch

Cut off the unused pins at the root. Shorten the 6 pins in use (common + 5 positions) so they cannot touch each other and bend them over by 90°; straight pins do not fit in the case. Solder on the 6-wire ribbon cable and insulate the joints with heat-shrink or hot glue.

Find out which wire belongs to which position by turning the switch with a multimeter on it, and mark the wire ends (1–5 lines with a marker in the photo).



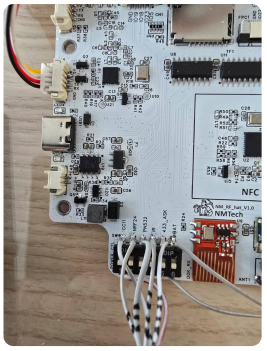


Prep 3 Solder the switch wires to the HAT

Set all 6 levers of the HAT's DIP switch to OFF. Solder the position wires of the rotary switch to the signal-side pads of the DIP in order: 1 CC1101, 2 NRF24, 3 PN532, 4 IR, 5 433_ASK. Connect the common wire to GND (the pads on the other side of the DIP are common GND).

Connect the middle pin of the slide switch to the signal pad of DIP 6 (BAT) and one outer pin to GND. With the switch on (I) the HAT switches the battery power on.

> Leave the DIP levers alone after soldering; if any of them is left ON, two modules will be on at once.



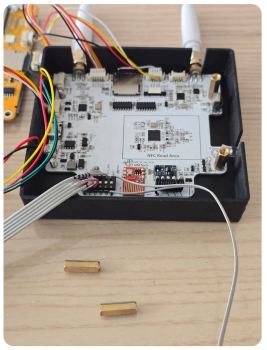


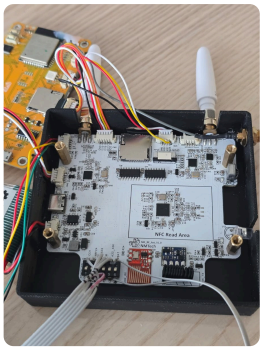
Assembly steps

Step 1 Screw the HAT into the case

Lower the HAT into the case from the front with its back (NFC area) against the back wall: the two SMA sockets go into the slots in the top wall, the IR LED into the IR TX slot in the right wall. Fit the four M3 × 6 screws from the back, through the HAT's holes, into the 15 mm standoffs you hold on the inside.

> The photos are from an earlier trial print; the inside details of the current case differ slightly.

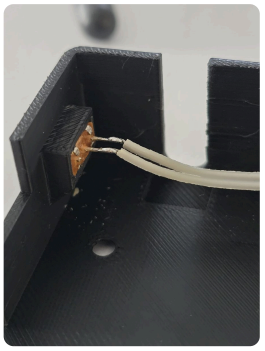




Step 2 Fit the slide switch into its holder

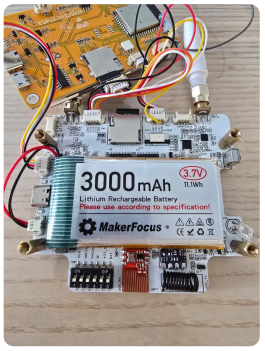
Slide the switch sideways into the holder at the top of the left wall from the inside, pin side last; its lever comes out through the window in the wall. Position I is on, O is off. Fix it with thin double-sided tape if needed.

> Photo from the NM-CYD-C5 version of the same case; this part is identical in both versions.



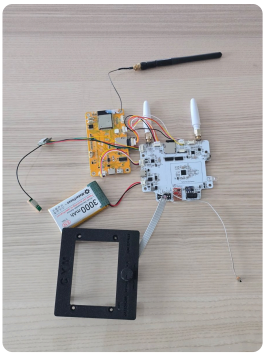
Step 3 Fit the battery

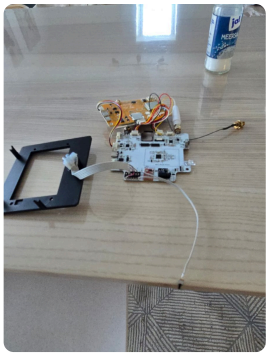
Stick the battery onto the front of the HAT, between the standoffs, with thin double-sided tape. Check the polarity with a multimeter before plugging it into the HAT's BAT socket; keep the switch off (O).



Step 4 Cable the CYD to the HAT

Plug the SD adapter into the CYD's microSD slot and connect its 5-wire cable to the "5P From CYD TF Extension" socket on the HAT. Join CYD P1 ↔ HAT IN and CYD CN1 ↔ HAT I2C with the two 4-wire cables. Lay the cables so that they will not be pinched when the CYD goes in.





Step 5 Fit the rotary switch to the front cover

Insert the switch into its seat from the inside of the front cover; the locking pin on its body sits in the Ø2.3 hole. Tighten the M9 nut from the front, inside the counterbore. Turn the switch to the CC1101 position (the one you marked as 1 in Prep 2, wired to DIP 1) and press the knob onto the shaft with its marker line pointing at the CC1101 label.

Try it: in each position the marker should point at the label of the module wired to it (CC1101, NRF24, PN532, IR, RF433). If not, pull the knob off and refit it, or check the wiring.



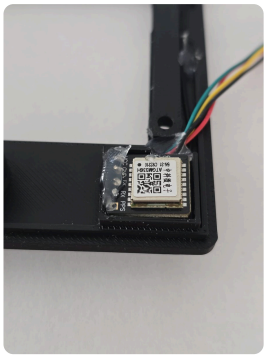


Step 6 Fit the GPS module and antenna into the front cover

Press the antenna plug onto the gold socket on the GPS module. Press the module into its pocket at the bottom left (seen from the front) with its LED side towards the cover and the LED over the GPS hole in the cover; the wires leave through the opening in the pocket. Place the antenna in its pocket at the top, ceramic side towards the cover. Fix with thin tape if needed.

Plug the GPS cable into the HAT's OUT / GNSS socket.

> Photo from the NM-CYD-C5 version of the same case; this part is identical in both versions.





Step 7 Test before closing

Turn the switch on (I). The CYD should power up and the screen backlight should come on. If the firmware is installed, turn the rotary switch and try the modules (Part 4); if not, install the firmware first.



Step 8 Seat the CYD, screw on the front cover

Lower the CYD onto the standoffs with the screen facing forward; no wire may be trapped between the board and the standoffs. Fit the front cover: the centering lip goes inside the case and the three tongues into the slots. Fit the four M3 × 7 screws from the front, crosswise, evenly and without overtightening.

> If the screen protector has a green pull tab, do not let it end up under the cover; remove the film or cut the tab off.





Step 9 Antennas, stylus, SD card and final check

Fit the two SMA antennas to the sockets on top: CC1101 on the left, NRF24 on the right. Push the stylus into its channel through the mouth at the lower-left corner; its head sits in the recess and is pulled out by the notch next to it. Insert the SD card through the SD slot in the top wall into the HAT's slot.

Use the HAT's USB-C socket (left wall, CHARGE) for charging; its two LEDs show in the recess. Use the CYD's MICRO or USB-C socket to install the firmware; the CHARGE socket carries no data.





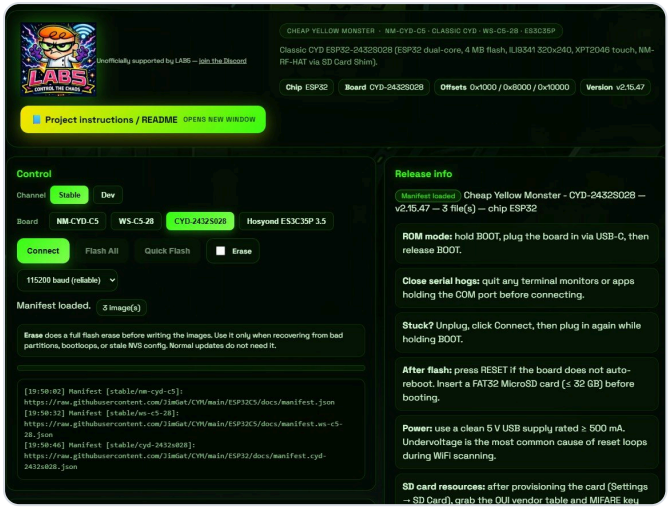
Part 4 – Firmware

Installing the firmware with the Web Flasher

You can install the firmware from the browser, without a USB driver or an IDE.

URL: https://jimgat.github.io/CYM/

1. Turn the slide switch off (O).
2. Open jimgat.github.io/CYM/ in Chrome or Edge and select the "CYD-2432S028" board.
3. Connect the CYD to the computer via its MICRO or USB-C socket, click Connect and then "Flash"; wait until the upload has finished.
4. When CYM starts, tap Enable on the Settings → Hardware Options → NM-RF-HAT screen (Status: ENABLED appears).
5. Select the module with the rotary switch (Part 3, rotary switch positions).



Web Flasher interface – "CYD-2432S028" board selected.

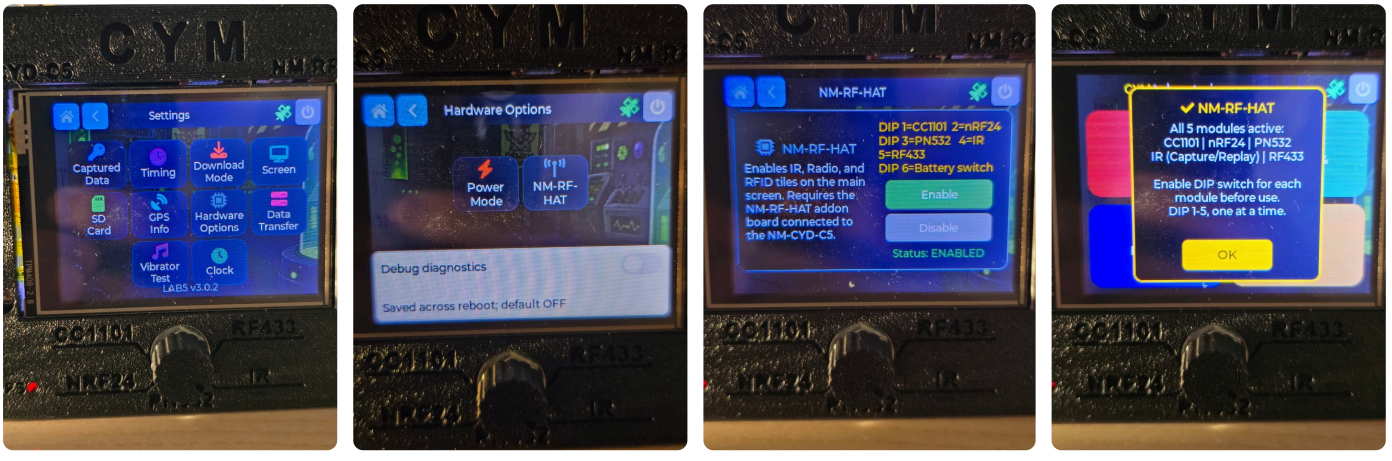
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> The Web Flasher only works in Chrome and Edge (it needs the Web Serial API). Safari and Firefox are not supported.

Enabling the NM-RF-HAT



Enabling the NM-RF-HAT: Settings → Hardware Options → NM-RF-HAT → Enable; then turn the rotary switch to the module you want. (Screens photographed on the NM-CYD-C5.)